

Capstone Project Analysis: Role of a Retail Store

Group 8 | IIM Calcutta | APAL02

1. PROJECT UNDERSTANDING (from PPT Analysis)

What Is This Project?

Your capstone is about building an **ML-enabled store-targeting tool for Lowe's** (a leading US home improvement retailer with ~1,759 stores and ~\$85B revenue). The core problem: Lowe's currently sets next-year store sales targets by simply spreading the total company plan using each store's 3-year historical sales share (a "peanut butter spread" approach). This ignores local market dynamics like population growth, income shifts, housing age, and competitive pressure changes.

The Proposed Solution

Replace that simplistic spreading with a data-driven approach that:

1. **Clusters stores into 5 roles:** High Growth, Growth, Neutral, Maintain, Defend — based on local demand and risk drivers
2. **Estimates store-level growth expectations** and seasonality patterns
3. **Allocates annual + monthly targets** via constrained optimization (targets must still sum to the division plan)
4. **Provides explainability:** driver-based explanations, peer benchmarks, planner overrides

POC Scope

The proof of concept focuses on the **West Division (~492 stores)** only. Scale chainwide after validation.

Success Measures

- Lower store-month plan error vs. current baseline spreading
 - Fewer extreme misses (% stores outside $\pm 10\text{-}15\%$)
 - Fewer mid-year re-targets/replanning
 - Division total plan met by construction (constraint)
-

2. DATASET OVERVIEW

What You Have

Dimension	Detail
Total Rows	269,412
Total Columns	195
Unique Stores	1,727
West Division Stores (POC)	492
Time Period	3 years (2023, 2024, 2025)
Granularity	Store-Week (52 weeks/year)
Weeks Per Store	~156 (52 weeks x 3 years)
Divisions	NORTH (94,380 rows), SOUTH (98,280), WEST (76,752)

Data Quality Notes

- **2023 weeks 27-52:** Actual Sales are missing for all 1,727 stores (looks like a data cutoff for the older year — only H1 2023 actuals available)
- **2025 data:** All 52 weeks present but actuals likely only through recent weeks
- **CYE % Household Income 250K Plus:** ~112K rows missing (41.6%) — a significant gap
- **2010 Total Housing Units:** ~22K rows missing (8.3%)
- **All other features:** Fully populated with zero or near-zero missing values

3. COMPLETE FEATURE INVENTORY

Your data has **195 columns** which fall into these categories:

Category	Count	Examples
Store Identity	7	Store ID, Name, District, Region, Division
Time	3	Year, Fiscal Week, Fiscal Week Ending Date
Sales/Performance (Target Variables)	4	Actual Sales USD, Plan Sales USD, Invoice Count, Avg Ticket
Store Physical	2	Sales Floor Size, Garden Ctr Size
Geography/Urbanicity	4	Urbanicity, CBSA Type, CBSA Metro Size, Area Sq Mi
Household Demographics	~18	

Category	Count	Examples
		Total Households (2010/2020/CYE), Income brackets, Density, Growth Rate
Housing Characteristics	~18	Housing Units, Ownership, Age of Housing (decade buckets), Home Values
Population Demographics	~25	Total Pop, Age, Education, Marital Status, Language, Veterans, Per Capita Income
Competition — Trade Area	18	One column per competitor within the store's trade area
Competition — 3mi radius	16	One column per competitor within 3 miles
Competition — 5mi radius	16	One column per competitor within 5 miles
Competition — 10mi radius	16	One column per competitor within 10 miles
Competition — 0-11mi radius	16	One column per competitor within 0-11 miles
Competition — 11-30mi radius	16	One column per competitor within 11-30 miles

4. RECOMMENDED FEATURES: 42 Features Across 7 Groups

After analyzing all 195 columns for relevance, data quality, variance, redundancy, and alignment with the PDF's priority weightage, here is my recommended feature set.

GROUP A: TARGET / DEPENDENT VARIABLES (4 features)

These are what you're predicting or benchmarking against — not inputs to the clustering model.

#	Feature	Role	Why
1	Actual Sales USD	Primary target	The actual weekly sales — this is what you're trying to allocate and predict
2	Plan Sales USD	Baseline benchmark	The current "peanut butter spread" plan — you need this to measure improvement
3	Invoice Count	Supporting metric	Transaction volume indicates store traffic patterns independent of ticket size
4	Avg Ticket	Supporting metric	Average transaction value helps distinguish high-value vs. volume stores

GROUP B: STORE PHYSICAL ATTRIBUTES (3 features)

#	Feature	Priority (PDF)	Why Include
5	Sales_Floor_Size	Y (no weight)	While the PDF notes fulfillment is largely unaffected by size, floor size still correlates with product assortment breadth and walk-in customer experience. Use as a control variable.
6	Garden_Ctr_Size	High	Explicitly called out as a "traffic driving category." Larger garden centers attract more footfall, especially in spring/summer seasonality. Directly impacts sales potential.
7	Urbanicity	High	7 levels from Metropolis to Remote. This is a master segmentation variable — it captures density, accessibility, customer behavior, and competitive dynamics in a single feature. Essential for clustering.

GROUP C: HOUSEHOLD & INCOME DEMOGRAPHICS (8 features)

The PDF lists 8 income bracket columns (Less than 25k through 250K+), all marked "High" priority. However, **using all 8 brackets creates multicollinearity** (they sum to ~100%). My recommendation:

#	Feature	Priority	Why Include
8	CYE_Total_Households	High	Total addressable market size — the single most important demand driver. More households = more potential customers.
9	CYE_Median_Household_Income	High	Best single summary of trade area purchasing power. Median is robust to outliers unlike mean/aggregate.
10	CYE_%_Household_Income_Less_than_25k	High	Captures the "value-conscious" segment. High percentage indicates a trade area skewed toward budget shoppers — affects product mix and ticket size.
11	CYE_%_Household_Income_75k_to_100K	High	The "sweet spot" for home improvement: households with enough income for projects but not wealthy enough to hire everything out. This is Lowe's core DIY customer.
12	CYE_%_Household_Income_150k_to_250k	High	Captures the "Pro/contractor" customer proxy — wealthier areas

#	Feature	Priority	Why Include
			tend to have more remodeling projects and higher-value purchases.

Why not all 8 brackets? Income distribution information is captured by the median plus the 3 strategic brackets above (low-end, core DIY, high-end). Including all 8 introduces near-perfect multicollinearity which destabilizes regression/clustering. If the model needs more income granularity, use PCA on all 8 to create 2-3 income components.

| 13 | **Compound_Annual_HH_Growth_Rate_2010_2020** | — | Not in PDF but present in CSV. Captures whether the trade area is growing or declining over a decade — a critical forward-looking indicator. Growing areas = "High Growth" role candidates. |

Dropped from this group: - *CYE_Aggregate_Family_Household_Income*: Marked "Low" priority in PDF. Aggregate income is redundant with Median + Total Households. -

CYE_Average_Mean_Household_Income: Redundant with Median; mean is more sensitive to outliers.

GROUP D: HOUSING CHARACTERISTICS (7 features)

#	Feature	Priority	Why Include
14	CYE_Total_Housing_Units	High	Total housing stock in trade area. Correlated with Total Households but captures vacant/new units too — important for renovation demand.
15	CYE_Median_Year_Housing_Unit_Built	High	Single best indicator of housing age. Older homes = more maintenance/repair needs (plumbing, electrical, roofing). Directly drives "Maintain" role identification.
16	CYE_%_Housing_Unit_Built_before_1949	High	Very old housing stock — highest maintenance demand. These homes need continuous repair.
17	CYE_%_Housing_Unit_Built_2000_to_2009	High	Recent construction — lower repair needs but higher "finishing" demand (landscaping, upgrades).
18	CYE_%_Housing_Unit_Built_after_2020	High	

#	Feature	Priority	Why Include
			Brand new construction — drives new-homeowner demand (appliances, fixtures, landscaping). High values signal "High Growth" areas.
19	CYE_Average_Mean_Length_of_Residence_Years	High	New move-ins have improvement needs; long-term residents have maintenance needs. A key behavioral segmentation variable per the PDF.

Why not all 8 decade buckets? Same multicollinearity concern as income brackets — they sum to ~100%. The three chosen brackets capture the strategic extremes (very old, recent, brand new), and the Median Year Built provides the central tendency. If needed, PCA on all decade buckets can extract 2-3 housing-age components.

| 20 | **CYE_%_Housing_Units_Owned** | Low (PDF) | Including despite "Low" PDF priority because home ownership is a fundamental driver of home improvement spending. Renters spend significantly less on home improvement. Use selectively. |

Dropped: - *CYE_%_House_Units_Rented*: Perfect inverse of % Owned — completely redundant. - *Individual decade buckets (1950s-1990s)*: Captured by Median Year Built + the 3 strategic buckets above.

GROUP E: POPULATION & DEMOGRAPHIC FEATURES (6 features)

#	Feature	Priority	Why Include
21	CYE_Total_Population	High	Raw market size. Complements Total Households — captures multi-person households and individual demand potential.
22	CYE_Median_Age_Total_Pop	High/ Medium	Age drives spending patterns: younger populations tend toward new homes/renovations; older populations toward maintenance/accessibility.
23	CYE_Veteran_Population	High	Veterans get a flat 10% discount at Lowe's — this directly impacts both traffic

#	Feature	Priority	Why Include
			and margin. A unique Lowe's-specific demand driver.
24	CYE_%_College_Graduate_Pop_25plus	—	Education level correlates with income and home ownership rates. Adds a dimension beyond just income for understanding the trade area profile.
25	CYE_%_Speak_only_English_at_Home_Pop_5plus	—	Language composition can indicate diverse communities with different shopping behaviors and product preferences.
26	CYE_Per_Capita_Income_Total_Pop	—	Complements Median Household Income by capturing individual-level purchasing power. Useful when household sizes vary significantly.

GROUP F: COMPETITIVE LANDSCAPE (11 features)

This is where careful selection matters most. The CSV has **98 competition columns** (18 competitors x 5-6 distance bands + trade area). Using all of them would drown the model in noise. Here is the strategic selection:

Major competitors (use Trade Area count — the most business-relevant geography):

#	Feature	Why Include
27	Nut_Cracker_Tools_Count_in_TradeArea	Described as "major competitor." Mean=3.45, only 10% zeros — highly prevalent and variable. This is likely the primary competitor (Home Depot analog).
28	Greek_Baths_Hardware_Count_in_TradeArea	"Major competitor." Mean=0.36 but up to 8 — significant in certain markets.
29	Bo_Jacks_Hardware_Count_in_TradeArea	"Major competitor." Mean=2.32, 19% zeros — widely present.
30	Dave_Gilmour_Tools_Count_in_TradeArea	"Major competitor." Mean=0.23 — lower presence but still strategically important.

#	Feature	Why Include
31	Janis_Jay_Store_Count_in_TradeArea	"Major competitor." Mean=1.41, 22% zeros — broad presence.

High-volume secondary competitors (use Trade Area count):

#	Feature	Why Include
32	Iggy_Pop_Hardware_Count_in_TradeArea	Mean=5.25, only 2% zeros — the most prevalent competitor. Despite being "small pockets," their sheer density matters.
33	Wallflowers_Depot_Count_in_TradeArea	Mean=4.68, only 1% zeros — nearly universal competitor presence.
34	Horn_Ok_Tools_Count_in_TradeArea	Mean=3.43, 15% zeros — widespread.
35	Mick_Js_Groceries_Count_in_TradeArea	Mean=2.47, 21% zeros — possibly a big-box/ grocery competitor that sells home improvement items.

Cannibalization (sister stores):

#	Feature	Why Include
36	Sister_Store_Count_in_TradeArea	Measures self-cannibalization. More sister stores = shared demand. Critical for understanding a store's true addressable market.

Aggregate competition pressure:

#	Feature (Engineered)	Why Include
37	Total_Competitor_Count_in_TradeArea	Sum of ALL competitor counts in trade area. A single composite measure of competitive intensity. Engineer this from the raw columns.

Why Trade Area counts and not 5mi/10mi? The PDF explicitly states: "Lowe's Trade Areas are derived through a proprietary process that accounts for approximately 70% of a location's annual sales." Trade area is the most business-meaningful geography. The 5mi/10mi radius features are crude proxies. Use one distance band (10mi) only as a robustness check, not as a primary feature.

Dropped: - *Thin_Lizzy_Hardware*: 100% zeros — this competitor has zero presence in the data. Completely useless. - *Bobs_Supplies*: 99% zeros — near-zero variance, no predictive value. - *All 3mi, 5mi, 0-11mi, 11-30mi bands*: Redundant with Trade Area. Would add 80+ columns of noise. - *Lumber_Jax, BB_King, Eddie_Vedder, Mile_One, Neil_Youngs, JJ_Cale*: Smaller competitors (77%, 30%, 35%, 10%, 43%, 52% zeros respectively). Include only Lumber_Jax or Neil_Youngs if initial model performance is weak and you want more granularity.

GROUP G: TIME & GEOGRAPHY IDENTIFIERS (5 features)

#	Feature	Why Include
38	Year	Captures year-over-year trends and macro shifts
39	Fiscal_Week	Captures seasonality (critical for monthly target allocation)
40	Region	Sub-division geography for regional effects within West Division
41	CBSA_Type	Metropolitan vs. Micropolitan — captures urban economic structure
42	CBSA_Metro_Size	Finer-grained metro classification

5A. LEAN STARTING SET: 20 Features for Phase 1

Rather than starting with all 42 features, a leaner initial set of ~20 features is recommended for the first iteration. This gives a healthier ratio of ~25 stores per feature (vs. ~12:1 with 42 features), making clustering significantly more stable. Run the full pipeline end-to-end with this set first, then add features in later iterations only if they improve model performance.

Why Start Lean?

- **Better clustering stability:** 492 stores ÷ 20 features = ~25:1 ratio (rule of thumb is 10-15:1 minimum)
- **Faster iteration:** Run the full pipeline (clustering → growth estimation → allocation → validation) quickly
- **Empirical feature selection:** Add features back based on whether they actually improve silhouette scores or reduce MAPE, rather than guessing upfront
- **Avoids curse of dimensionality:** Distance metrics work better in lower-dimensional space
- **Simpler explainability:** Easier to present 20 well-chosen features to business stakeholders

The 20-Feature Starting Set

#	Feature	Group	Why It Made the Cut
1	Actual Sales USD	A: Target	Primary target variable
2	Plan Sales USD	A: Target	Baseline benchmark for measuring improvement
3	Invoice Count	A: Target	Traffic patterns independent of ticket size
4	Avg Ticket	A: Target	Distinguishes high-value vs. volume stores

#	Feature	Group	Why It Made the Cut
5	Urbanicity	B: Store Physical	Master segmentation variable — 7 levels capturing density, accessibility, behavior
6	Garden_Ctr_Size	B: Store Physical	Explicitly called out as "traffic driving category" in PDF
7	CYE_Total_Households	C: Household	Single most important demand driver — total addressable market size
8	CYE_Median_Household_Income	C: Household	Best single summary of purchasing power (replaces all 8 income brackets)
9	Compound_Annual_HH_Growth_Rate_2010_2020	C: Household	Forward-looking indicator — growing vs. declining trade areas
10	CYE_Total_Housing_Units	D: Housing	Total housing stock including vacant/new — drives renovation demand
11	CYE_Median_Year_Housing_Unit_Built	D: Housing	Single best indicator of housing age (replaces all decade buckets)
12	CYE_Average_Mean_Length_of_Residence_Years	D: Housing	New move-ins vs. long-term residents — key behavioral variable
13	CYE_Total_Population	E: Population	Raw market size, complements Total Households
14	CYE_Veteran_Population	E: Population	Lowe's-specific: 10% veteran discount directly impacts traffic and margin
15	Nut_Cracker_Tools_Count_in_TradeArea	F: Competition	Primary competitor (Home Depot analog). Mean=3.45, only 10% zeros
16	Wallflowers_Depot_Count_in_TradeArea		

#	Feature	Group	Why It Made the Cut
		F: Competition	Most prevalent competitor. Mean=4.68, only 1% zeros
17	Sister_Store_Count_in_TradeArea	F: Competition	Self-cannibalization — critical for true addressable market
18	Total_Competitor_Count_in_TradeArea	F: Competition	Engineered: composite measure of competitive intensity
19	Year	G: Time	Year-over-year trends
20	Fiscal_Week	G: Time	Seasonality (critical for monthly allocation)

What's Deferred to Phase 2 (Add Back If Needed)

Deferred Feature	Original Group	When to Add Back
Sales_Floor_Size	B	If clustering doesn't distinguish store capacity well
CYE_%_HH_Income_Less_than_25k	C	If clusters don't separate value vs. premium markets
CYE_%_HH_Income_75k_to_100K	C	If core DIY segment isn't captured by median income alone
CYE_%_HH_Income_150k_to_250k	C	If Pro/contractor demand isn't differentiated
CYE_%_Housing_Unit_Built_before_1949	D	If housing age median alone misses the "very old homes" signal
CYE_%_Housing_Unit_Built_2000_to_2009	D	If recent construction demand isn't captured
CYE_%_Housing_Unit_Built_after_2020	D	If new-build areas aren't identified as High Growth
CYE_%_Housing_Units_Owned	D	If ownership rate matters beyond income proxy
CYE_Median_Age_Total_Pop	E	If age-driven spending patterns aren't captured

Deferred Feature	Original Group	When to Add Back
CYE_%_College_Graduate_Pop_25plus	E	If education adds predictive power beyond income
CYE_%_Speak_only_English_at_Home	E	If community diversity affects shopping behavior
CYE_Per_Capita_Income_Total_Pop	E	If individual income adds signal beyond household income
Greek_Baths_Hardware, Bo_Jacks_Hardware, Dave_Gilmour_Tools, Janis_Jay_Store	F	If top-2 competitors + total count don't capture competitive dynamics
Iggy_Pop_Hardware, Horn_Ok_Tools, Mick_Js_Groceries	F	If secondary competitor density matters
Region, CBSA_Type, CBSA_Metro_Size	G	If geographic effects aren't captured by Urbanicity alone

How to Decide What to Add Back

1. **Run clustering with 20 features** → check silhouette score and business interpretability
2. **Run growth model with 20 features** → check MAPE and feature importance (SHAP)
3. **Add one group at a time** (e.g., all deferred income brackets) → re-run → check if metrics improve
4. **Keep only features that improve metrics by >1%** — otherwise they're adding noise

5B. FULL SET: 42 FEATURES RECOMMENDED

Group	Count	Features
A: Target Variables	4	Actual Sales, Plan Sales, Invoice Count, Avg Ticket
B: Store Physical	3	Sales Floor Size, Garden Ctr Size, Urbanicity
C: Household & Income	6	Total Households, Median Income, 3 income brackets, HH Growth Rate
D: Housing	7	Total Housing Units, Median Year Built, 3 age buckets, Residence Length, % Owned
E: Population	6	Total Pop, Median Age, Veterans, College %, English %, Per Capita Income
F: Competition	11	

Group	Count	Features
		5 major competitors, 4 high-volume secondary, Sister Stores, Total Competitor (engineered)
G: Time & Geography	5	Year, Fiscal Week, Region, CBSA Type, CBSA Metro Size
TOTAL	42	Down from 195 raw columns (78% reduction)

What You're Dropping and Why

Dropped Category	Count	Reason
Redundant income brackets	5	Multicollinearity — captured by Median + 3 strategic brackets
Redundant housing decade buckets	5	Multicollinearity — captured by Median Year + 3 strategic buckets
% Rented	1	Perfect inverse of % Owned
Aggregate/Mean income	2	Redundant with Median Household Income
Zero-variance competitors	2	Thin Lizzy (100% zeros), Bobs Supplies (99% zeros)
Redundant distance bands (3mi/5mi/10mi/0-11mi/11-30mi)	~80	Trade Area is the gold standard per Lowe's methodology
Minor competitors	6	Low presence, high zeros — keep as fallback
Store identity columns	4	Real Store ID, Store Name, District (use Region instead)
Historical demographics (2010, 2020 snapshots)	~10	CYE (current year estimates) are more relevant; growth rate captures trend
Remaining low-signal population features	~20	Marital status, rural %, various language breakdowns — weak home improvement drivers

6. PROJECT PLAN: How to Execute This Capstone

Phase 0: Setup & Data Preparation (Weeks 1-2)

Goal: Clean data, engineer features, create the analytical dataset.

Tasks: 1. **Filter to West Division** — reduce from 1,727 to 492 stores 2. **Handle missing data** — 2023 H2 actuals are missing; use 2024 as primary training year with 2023 H1 as supplementary. Impute `CYE_%_Household_Income_250K_Plus` (41.6% missing) — either drop it or impute with median by Urbanicity 3. **Aggregate weekly to store-level annual summaries** — for clustering,

you need one row per store with: Annual Sales (TY, LY), YoY growth %, seasonality profile (e.g., peak week sales / average week sales), average weekly invoice count, average ticket 4. **Engineer features**: Total_Competitor_Count_in_TradeArea (sum of all competitor counts), YoY_Sales_Growth_Pct, Seasonality_Index, Plan_vs_Actual_Ratio (how well current planning works for each store) 5. **Normalize/standardize** all numerical features for clustering (StandardScaler or RobustScaler) 6. **Create a store-level master table**: one row per store, 42 features

Deliverable: Clean, analysis-ready dataset (492 stores x ~45 features)

Phase 1: Exploratory Data Analysis (Week 2-3)

Goal: Understand distributions, correlations, and natural groupings.

Tasks: 1. **Univariate analysis** — distribution of each feature, outlier detection 2. **Correlation matrix** — identify highly correlated pairs (>0.85) for potential further reduction 3. **Plan vs. Actual analysis** — quantify how bad the current "peanut butter" approach is: by store, by month, by urbanicity. This becomes your baseline 4. **Geographic visualization** — map stores colored by current plan accuracy, income levels, competition density 5. **Seasonality analysis** — identify seasonal patterns by store cluster (rural vs. metro stores may peak differently)

Deliverable: EDA report with key insights, baseline error quantification

Phase 2: Store Role Clustering (Weeks 3-5)

Goal: Cluster 492 stores into 5 meaningful roles.

Tasks: 1. **Feature selection for clustering** — use only the demand/market features (Groups B-F, not sales targets). Consider PCA for dimensionality reduction 2. **Try multiple clustering approaches**: - **K-Means** ($k=5$) — simple, interpretable baseline - **Hierarchical clustering** — reveals natural groupings, dendrogram visualization - **Gaussian Mixture Models (GMM)** — softer boundaries, probability-based assignment 3. **Validate clusters** — Silhouette score, Davies-Bouldin index, and most importantly: **business interpretability**. Each cluster should map clearly to one of the 5 roles 4. **Label clusters** — based on their feature profiles, assign meaningful role names: High Growth (growing area, new housing, high income), Growth (moderate positive signals), Neutral (stable market), Maintain (aging housing, steady population), Defend (declining market, high competition) 5. **Profile each role** — create a "persona" for each cluster with key driver values

Deliverable: 5 well-defined store roles with business-meaningful profiles

Phase 3: Growth Estimation & Target Allocation (Weeks 5-7)

Goal: Build the constrained allocation model.

Tasks: 1. **Estimate store-level growth rates** — use role-specific regression models (or gradient boosting) to predict next-year sales for each store based on its features and historical trajectory 2. **Seasonality modeling** — for each role, learn monthly/weekly sales phasing (e.g., Fourier decomposition or simple seasonal indices) 3. **Constrained optimization** — formulate the allocation problem: maximize target accuracy per store SUBJECT TO all store targets summing to the division plan. This is a constrained least-squares or proportional scaling problem 4. **Monthly target allocation** — apply role-specific seasonality profiles to annual targets to generate monthly targets 5. **Compare to baseline** — measure Plan Error (your allocation vs. current allocation) against actuals

Deliverable: Store-level annual + monthly targets for all 492 West Division stores

Phase 4: Explainability & Planner Tool (Weeks 7-9)

Goal: Make the model transparent and usable.

Tasks: 1. **SHAP/Feature importance** — for each store, show which features drove its role assignment and growth estimate 2. **Peer benchmarking** — for each store, show its closest peers (same role, similar features) and their performance 3. **Risk flagging** — identify stores with high uncertainty in predictions (wide confidence intervals) for planner review 4. **Build a simple front-end** — a Streamlit or Gradio app where planners can: input a division growth target (%), see how it allocates to each store by role, drill into any store to see driver explanations, override and see downstream impact 5. **What-if scenarios** — allow planners to change the growth target and see real-time reallocation

Deliverable: Interactive planner tool with explainability

Phase 5: Validation & Presentation (Weeks 9-11)

Goal: Prove the approach works better than the baseline.

Tasks: 1. **Back-testing** — use 2023 features to predict 2024 actuals, compare your allocation vs. the actual 2024 plan 2. **Key metrics:** Mean Absolute % Error (MAPE) by store, % of stores within $\pm 10\%$ of actual, % of stores within $\pm 15\%$ of actual, reduction in extreme misses vs. baseline 3. **Role stability analysis** — do stores stay in the same role when you re-run with different time periods? Stable roles = trustworthy model 4. **Prepare capstone presentation** — results, methodology, business impact, limitations, scale-up plan 5. **Document the full pipeline** — code, data dictionary, model cards

Deliverable: Final capstone report + presentation

Timeline Summary

Phase	Weeks	Key Output
0: Data Prep	1-2	Clean dataset (492 stores x 45 features)
1: EDA	2-3	Baseline error analysis, key insights
2: Clustering	3-5	5 store roles with profiles
3: Allocation Model	5-7	Constrained annual + monthly targets
4: Explainability & Tool	7-9	Streamlit planner app
5: Validation & Presentation	9-11	Back-test results, final report

Recommended Tech Stack

Component	Tool	Why
Data Processing	Python (pandas, numpy)	Standard, your team likely knows it
Clustering	scikit-learn (KMeans, GMM, Hierarchical)	Well-documented, easy to compare
Growth Modeling	XGBoost or LightGBM	Handles mixed feature types, built-in feature importance
Constrained Optimization	scipy.optimize or cvxpy	For constrained target allocation
Explainability	SHAP	Industry standard for ML explainability
Seasonality	statsmodels (seasonal decompose) or Prophet	Robust seasonal decomposition
Front-end	Streamlit	Fast prototyping, interactive, Python-native
Visualization	plotly, seaborn, matplotlib	Interactive charts for the planner tool

Key Risks & Mitigations

Risk	Impact	Mitigation
2023 H2 missing actuals	Reduces training data	Use 2024 as primary year; 2023 H1 as supplementary
Income 250K+ column 41.6% missing	Loss of high-income signal	Impute by Urbanicity median; or drop and rely on 150K-250K bracket

Risk	Impact	Mitigation
Cluster instability	Roles change with minor data changes	Use ensemble clustering; validate with multiple methods
Overfitting on 492 stores	Small sample for ML	Use regularization; prefer simpler models; cross-validate rigorously
Seasonality varies by store	One-size-fits-all monthly allocation fails	Use role-specific seasonal profiles, not chain-level

Document prepared for Group 8 Capstone / April 2026